higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANOTECHNICS N6

1 AUGUST 2019

This marking guideline consists of 10 pages.

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QUESTION 1: BRAKES

1.1 $10x + 9\left(\frac{x}{1,5}\right) = 288$

$10x + 6x = 288$

$x = 18°$

$\therefore \theta = \frac{18°}{2} = 9°$ (3)

1.2 $P = \frac{2\pi NT}{60}$

$150 \times 10^3 = \frac{2\pi(250)T}{60}$

$\therefore T = 5,73 \ kNm$ (2)

1.3 $\frac{T_1}{T_2} = \left[\frac{1 + \mu tan\theta}{1 - \mu tan\theta}\right]^n$

$\frac{T_1}{T_2} = \left[\frac{1 + 0,2\tan 9}{1 - 0,2\tan 9}\right]^{10}$

$T_1 = 1,885 \ T_2$ (3)

1.4 $T = (T_1 - T_2)R$

$5\ 730 = (1,885\ T_2 - T_2)0,4$

$T_2 = 16,186 \ kN$

$T_1 = 30,51 \ kN$

$T_1 \times 150 = T_2 \times 100 + F \times 600$

$30\ 510 \times 150 = 16\ 186 \times 100 + F \times 600$

$F = 4,93 \ kN$ (6)

[14]

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QUESTION 2: FRICTION CLUTCH

2.1 $h = w\sin\theta$

$= 80{,}9\sin 18°$

$= 25\ \ mm$

$D = d + 2h$

$= 140 + 2(25)$

$= 190\ \ mm$ (3)

2.2 $T = \dfrac{60P}{2\pi N}$

$= \dfrac{60(15 \times 10^3)}{2\pi(1\ \ 440)}$

$= 99{,}472\ \ Nm$

$R_f = \dfrac{R+r}{2} = \dfrac{95+70}{2} = 82{,}5\ \ mm$

$T = \dfrac{\mu F_A \times R_f}{sin\theta}$

$99{,}472 = \dfrac{(0{,}3)F_A \times (82{,}5\ \times\ 10^3)}{\sin 18°}$

$F_A = 1\ \ 241{,}961\ \ N$

$F_{engage} = F_A\left(1 + \dfrac{\mu}{tan\theta}\right)$

$= 1\ \ 241{,}961\left(1 + \dfrac{0{,}3}{\tan 18°}\right)$

$= 2\ \ 388{,}67\ \ N$ (7)

2.3 $P_{MAX} = \dfrac{F_A}{2\pi(R-r)r}$

$= \dfrac{1\ \ 241{,}961}{2\pi(0{,}095-0{,}07)0{,}07}$

$= 112{,}95\ \ kPa$ (2)

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QUESTION 3: LINE SHAFTS

3.1 $$P=\frac{2\pi N_A T_A}{60}$$

$$40\times10^3=\frac{2\pi(1\ 000)T_A}{60}$$

$$T_A=381{,}972\ \ Nm$$ (2)

3.2 $$D_A N_A=D_B N_B$$

$$(140)(1\ 000)=(280)N_B$$

$$N_B=500\ \ r/min$$

$$P=\frac{2\pi N_B T_B}{60}$$

$$40\times10^3=\frac{2\pi(500)T_B}{60}$$

$$T_B=763{,}944\ \ Nm$$ (4)

3.3 $$F_t=\frac{2\times T}{PCD}$$

$$=\frac{2\times381{,}972}{0{,}14}$$

$$=5{,}457\ \ kN$$ (1)

3.4 $$\frac{T_1}{T_2}=e^{\mu\theta}$$

$$\frac{T_1}{T_2}=e^{(0{,}43)(\pi)}$$

$$T_1=3{,}861\ \ T_2$$

$$T=(T_1-\ T_2)$$

$$763{,}944\ =(3{,}861\ \ T_2-\ T_2)0{,}175$$

(5)

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$T_2 = 1\ \ 525{,}828\ \ N$

$T_1 = 5\ \ 891{,}222\ \ N$

3.5 $\Sigma CWM = \Sigma ACWM$

$(882{,}9 \times 300) + (5\ \ 457 \sin 30° \times 800) + (196{,}2 \times 1\ \ 250) = R_R \times 1\ \ 000$

$R_R = 2\ \ 696{,}92\ \ N$

$\Sigma CWM = \Sigma ACWM$

$(R_L \times 1\ \ 000) + (196{,}2 \times 250) = (882{,}9 \times 700) + (5\ \ 457 \sin 30° \times 200)$

$R_L = 1\ \ 114{,}68\ \ N$ (4)

3.6 $\Sigma CWM = \Sigma ACWM$

$(5\ \ 457 \cos 30°)(800) + (R_R \times 1\ \ 000) = (1\ \ 525{,}828 + 5\ \ 891{,}222)(1\ \ 250)$

$R_R = 5\ \ 490{,}592\ \ N$

$\Sigma CWM = \Sigma ACWM$

$(1\ \ 525{,}828 + 5\ \ 891{,}222)(250) + (R_L \times 1\ \ 000) + (5\ \ 457 \cos 30°)(200) = 0$

$R_L = -2\ \ 799{,}443\ \ N$ (4)

[20]

QUESTION 4: FLYWHEEL

4.1 $$\propto = \frac{w_2 - w_1}{t}$$

$$= \frac{47,124 - 0}{8}$$

$$= 5,89 \ \ rad/s^2$$

$$w_1 = \frac{2\pi N}{60}$$

$$= \frac{2\pi(450)}{60}$$

$$= 47,124 \ \ rad/s$$ (3)

4.2 $$T = I \propto$$

$$= (45)(5,89)$$

$$= 265,05 \ \ Nm$$

$$I = mk^2$$

$$= (500)(0,3)^2$$

$$= 4,5 \ \ kg.m^2$$ (3)

4.3 $$E_k = \frac{1}{2} Iw^2$$

$$= \frac{1}{2}(45)(47,124)^2$$

$$= 49,965 \ \ kJ$$ (2)

4.4 $$V = wr$$

$$= 47,124 \times 0,6$$

$$= 28,274 \ \ m/s$$ (2)

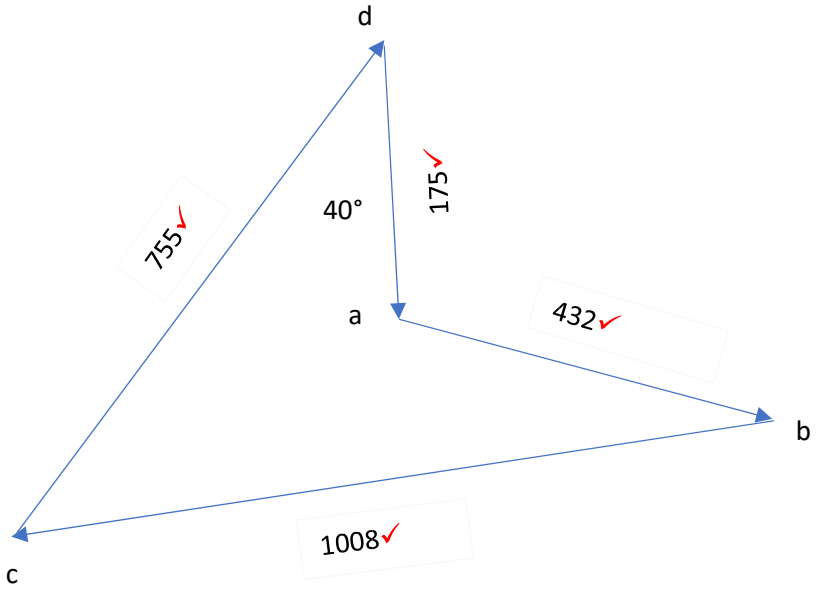
[10]

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QUESTION 5: BALANCING

5.1

	m(kg)	r(mm)	mr(kg.mm)	l ¿	$mr\,l$ ¿
A	10	70	700√	-0,25	-175√
B	8	90	720√	0,6	432√
C	12	60	720√	1,4	1 008√
x	-	-	R_x	0	0
y	-	-	R_y	1,2√	$1{,}2R_y$ √



$755 = 1{,}2\,R_y$

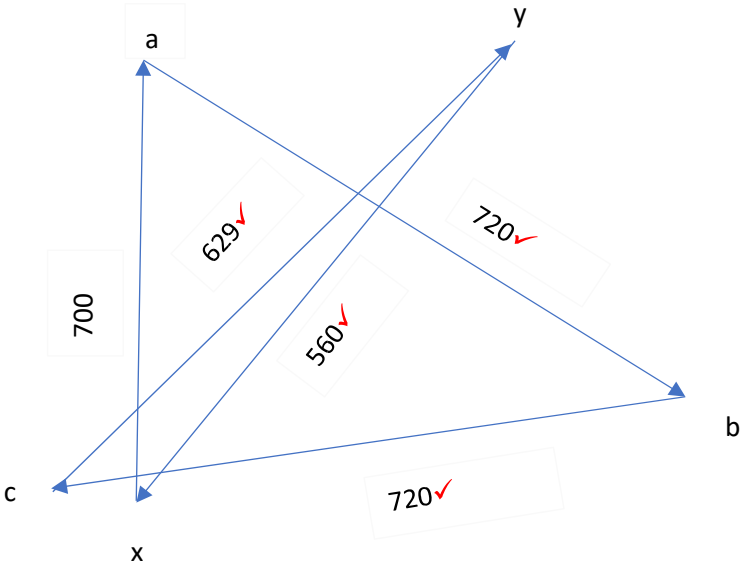
$R_y = 629{,}167 \;\; kg\,.\,mm$

$$dynamic \;\; reaction \;\; at \;\; y = \frac{629{,}167}{1\;\;000}\left(\frac{2\pi N}{60}\right)^2$$

$$= \frac{629{,}167}{1\;\;000}\left[\frac{2\pi(120)}{60}\right]^2$$

$$= 99{,}354 \;\; N \qquad (11)$$

5.2



$R_x = 560 \ \ kg.mm$

$dynamic \ \ reaction \ \ at \ \ x = \frac{560}{1\ \ 000}\left(\frac{2\pi N}{60}\right)^2$ ✓

$= \frac{560}{1\ \ 000}\left(\frac{2\pi\ 120}{60}\right)^2$

$= 88{,}432 \ \ N$ (9)

[20]

QUESTION 6: DYNAMICS

6.1 6.1.1 $N_f = 1{,}8 \times 10^3$

$= 7\ \ 063{,}2$

$N_R = 1{,}8 \times 10^3$

$= 10\ \ 594{,}8 \ \ N$

$F_a = \mu(mg)\frac{h}{d}$

$W_1 = 0{,}4(1\ \ 800 \times 9{,}81)\frac{0{,}7}{3{,}2}$

$W_1 = 1\ \ 545{,}075 \ \ N$

$\therefore Normal \ \ reaction \ \ rear \ \ wheel = N_R - W_1$

$= 10\ \ 594{,}8 - 1\ \ 5$ (6)

$= 9\ 049{,}725\ N$

6.1.2 $F_a = \mu(N_f + W_2)\frac{h}{d}$

$\therefore W_2 = \mu(N_f + W_2)\frac{h}{d}$

$W_2 = 0{,}4(7\ 063{,}2 + W_2)\frac{0{,}7}{3{,}2}$

$W_2 = 2\ 825{,}28 + 87{,}5 \times 10^{-3} W_2$

$W_2 = 3\ 096{,}197\ N$

$Normal\ reaction\ rear\ wheel = N_R - W_1$

$= 10\ 594{,}8 - 3\ 096{,}1$

$= 7\ 498{,}603\ N$ (4)

6.2 $W_3 = \mu(N_R - W_3)\frac{h}{d}$

$W_3 = 0{,}4(10\ 594{,}8 - W_3)\frac{0{,}7}{3{,}2}$

$W_3 = 4\ 237{,}92 - 87{,}5 \times 10^{-3} W_3$

$W_3 = 3\ 896{,}938\ N$

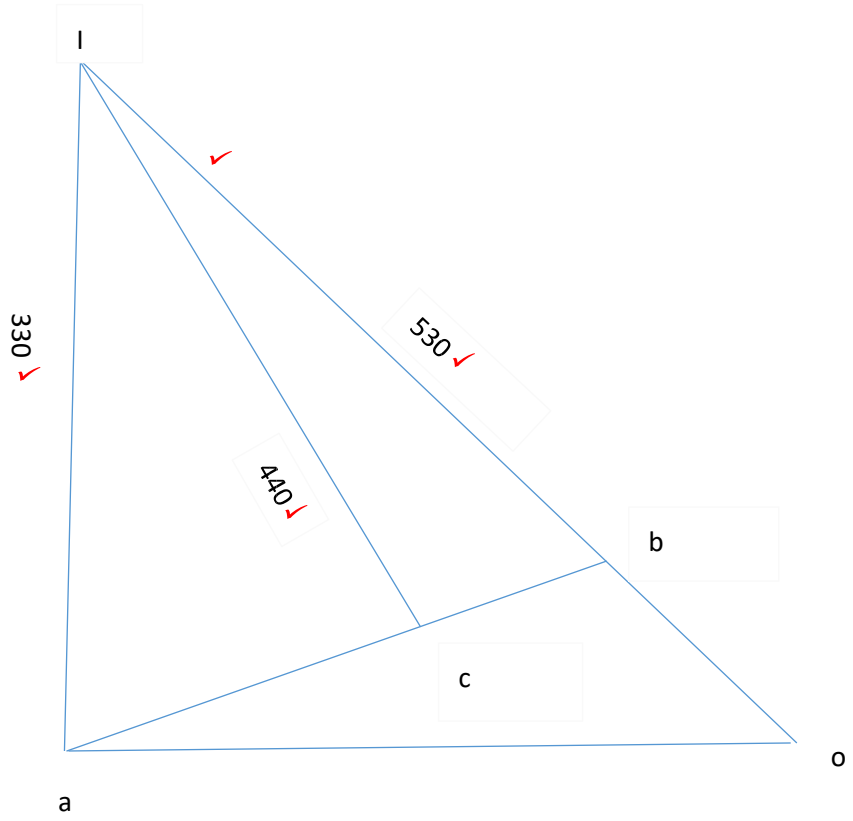
$Normal\ reaction\ on\ front\ wheel = N_f + W_3$

$= 7\ 063{,}2 + 3\ 896{,}938$

$= 10\ 960{,}138\ N$ (4)

[14]

QUESTION 7: KINEMATICS



$V_B = wr$

$= \left(\frac{2\pi(100)}{60}\right)(0,12)$

$= 1,257 \ \ m/s$

$V_A = \frac{V_B}{I_B} \times I_A$

$\text{¿} \frac{1,257}{530} \times 330$

$= 0,783 \ \ m/s$

$V_C = \frac{V_B}{I_B} \times I_C$ **OR** $V_C = \frac{V_A}{I_A} \times I_C$

$\text{¿} \frac{1,257}{530} \times 440$ $\text{¿} \frac{0,783}{330} \times 440$

$= 1,044 \ \ m/s$ $= 1,044 \ \ m/s$ **[10]**

TOTAL: 100

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